Chapter 7

Discrete random variables and binomiale

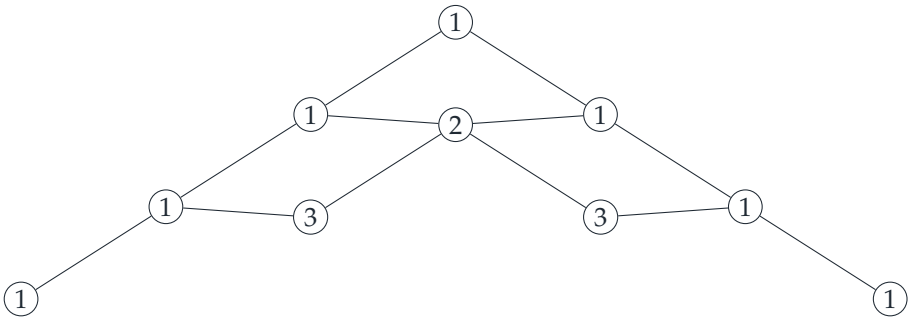
Competences: define a distribution, calculate and interpret a expectation, recognize and use a binomial distribution

7.1 Reference course

A discreet random variable X associates a real number with each outcome. His law gives the values x_i and Probability $p_i = P(X = x_i)$, $p_i \geq 0$ and $\sum p_i = 1$.

Key ideas

Expectation is $E(X) = \sum_i x_i p_i$. It represents the long-term average value when experience is repeated under the same conditions; it is not necessarily a value actually taken by X.

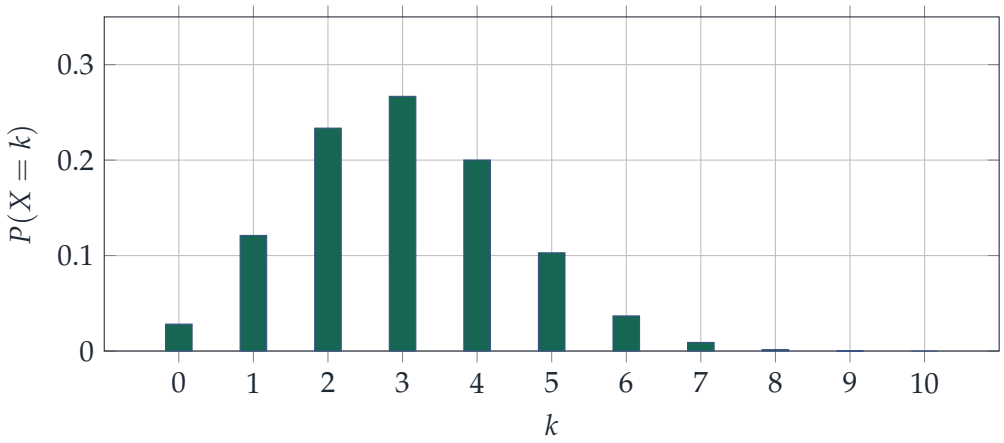


Key ideas

X follows a binomial distribution $\mathcal{B}(n, p)$ where X counts the number of successes in n test Bernoulli independent, similarly likely to succeed p. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad E(X) = np.$$

The coefficient $\binom{n}{k}$ count the number of paths containing exactly k success among n Tests. $n \leq 10$, it can be read in Pascal's triangle.



The diagram represents approximately $\mathcal{B}(10, 0{,}3)$; its centre is close to $E(X) = 3$.

Meeting 47 — Expectance of a random variable

Discrete random variables

06/04/2027 • B : 2 h – C : 1 h

Objectives of the session

· Define a law.

· Calculate a expectation.

· Interpret a theoretical average.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 8 - x + 2$. $2x + 10$
2. Provide $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3
3. Calculate $\binom{5}{2}$. 10
4. Calculate the expectation of X constant equal to 8. 8
5. Translate $X \geq 1$ by a supplementary. $1 - P(X = 0)$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: define a law and calculate a expectation.

Research activity

A game costs 2 euros. The gross gain is 0, 3 or 10 euros with probability 0 ,6, 0,3, 0,1. Building the law of net gain X, calculate $E(X)$ and decide if the game is favorable to the player.

Detailed correction · Professor edition

X value $-2, 1, 8$. $E(X) = -1{,}2 + 0{,}3 + 0{,}8 = -0{,}$Euro 1 : unfavorable game on average. the data that triggers the choice of method.

Trace written

The law of X associated with each value x_i probability p_i. Expectation $E(X) = \sum x_i p_i$ is an average long-term theory.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

X value $-2, 1, 8$. $E(X) = -1{,}2 + 0{,}3 + 0{,}8 = -0{,}1$ euro: on average unfavourable game.

Training · essential

1. Justify that the number of successes among 8 independent trials, with $p = 0{,}3$, follows a law binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduct from it $P(X \geq 1)$ and calculate $E(X)$.

Detailed correction · Professor edition

$X \sim \mathcal{B}(8, 0{,}3)$. $P(X = 2) = \binom{8}{2} 0{,}3^2 0{,}7^6$, $P(X = 0) = 0{,}7^8$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{24}{10}$.

Reference in the manual

Supplements: Calao, course pp. 154-159; exercises p. 160, no. 1-5.
session is identified below by its page and number; the reference also allows you to find
the figures and digital resources of the manual.

Manuel Calao · p. 160, exercise 2

A game consists of launching a well balanced dice. If the result is:

· 1, 2 or 3, then you lose 2 points;

· 3 or 4, so you don't lose or gain anything;

· 6, then we win 3 points.

Set G the random variable that, at each throw, combines the player's (positive or negative) gain.

1. Complete the table that represents the probability distribution of the random variable G.
2. Calculate $E(G)$. What can we notice?

Detailed correction · Professor edition

The printed text shows result 3 in two lines; the coherent partition is 1, 2, 3 for -2, 4, 5 for 0, and 6 for 3. $/2$, $1/3$ and $1/6$, then $E(G) = -2 \times 1/2 + 3 \times 1/6 = -0.5$. The game is unfavourable to the player on average.

Personal work

Resuming Exercise 2 with the following data:
355 and rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and
indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 48 — Binomial distribution: recognizing a situation

Discrete random variables

08/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Recognize a Bernoulli scheme.

· Identify n and p.

· Define X.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 180. 9

2. Calculate $\binom{5}{2}$. 10

3. Calculate the expectation of X constant equal to 9. 9

4. Translate $X \geq 1$ by a supplementary. $1 - P(X = 0)$

5. Provide $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reworded: recognize a bernoulli scheme and identify n and p.

Research activity

A customer accepts an offer with probability 0 ,25. An independent adviser contacts eight Clients. Define X, justify the binomial distribution and identify n and p.

Detailed correction · Professor edition

X counts acceptances; repetition of 8 identical and independent tests, success of Probability 0.25: $X \sim \mathcal{B}(8, 0{,}25)$. Verbate the data that triggers the choice of method.

Trace written

$X \sim \mathcal{B}(n, p)$ if X counts the successes in n independent tests of the same probability p.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

X counts acceptances; repetition of 8 identical and independent tests, success of Probability 0.25: $X \sim \mathcal{B}(8, 0{,}25)$.

Training · essential

1. Justify that the number of successes among 9 independent trials, with $p = 0{,}3$, follows a law binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduct from it $P(X \geq 1)$ and calculate $E(X)$.

Detailed correction · Professor edition

$X \sim \mathcal{B}(9, 0{,}3)$. $P(X = 2) = \binom{9}{2} 0{,}3^2 0{,}7^7$, $P(X = 0) = 0{,}7^9$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{27}{10}$.

Reference in the manual

Supplements: Calao, courses p. 154-159; exercises p. 160, n° 5, and p. 162, n° 14-20. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 163, exercise 22

The mayor of a city has decided to hold a referendum on the possible development of a park.
The results are as follows: 63% for and 37% against.
The number of participants is large enough to be able to assimilate this draw at random and with two ballot papers.
The random variable is considered X which, in any draw of two ballots, combines the number "For" bulletins.

1. What is the law followed by X ? Determine its parameters.
2. Calculate the probability that, in such a sampling, one has: none "For"; exactly one " For."
3. Deduce from question 2 the probability that, in such a sampling, there is at most one "For ».

Detailed correction · Professor edition

$X \sim \mathcal{B}(2; 0{,}63)$. We have $P(X = 0) = 0{,}37^2 = 0{,}1369$ and $P(X = 1) = 2 \times 0{,}63 \times 0{,}37 = 0{,}4662$.
$P(X \leqslant 1) = 0{,}6031$.

Personal work

Resuming Exercise 3 with the following data:
360 and rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 49 Pascal triangle and binomial coefficients

Discrete random variables 27/04/2027 • B : 2 h – C : 1 h

Objectives of the session

· Build Pascal's triangle.

· Read $\binom{n}{k}$.

· Interpret a coefficient.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase. $1+\frac{10}{100}=1,10$
2. Calculate the expectation of X constant equal to 3. 3
3. Translate $X \geq 1$ by a supplementary. $1-P(X=0)$
4. Provide $E(X)$ for $X \sim \mathcal{B}(10,0{,}3)$. 3
5. Calculate $\binom{5}{2}$. 10

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: build the pascal triangle and read $\binom{n}{k}$.

Research activity

Build Lines $n=0$ à $n=6$ of Pascal's triangle. Explain the calculation of the number 15 of the line 6, then read $\binom{6}{2}$ and $\binom{6}{4}$.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Detailed correction · Professor edition

Each inner number is the sum of the two numbers above. $\binom{6}{2}=\binom{6}{4}=15$. Doing verbalize the data that triggers the choice of method.

Trace written

Pascal's triangle checks $\binom{n}{k}=\binom{n-1}{k-1}+\binom{n-1}{k}$. The edges are worth 1.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

Each inner number is the sum of the two numbers above. $\binom{6}{2} = \binom{6}{4} = 15$.

Training · essential

1. Justify that the number of successes among 10 independent trials, with $p = 0{,}3$, follows a law binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduct from it $P(X \geq 1)$ and calculate $E(X)$.

Detailed correction · Professor edition

$X \sim \mathcal{B}(10, 0{,}3)$. $P(X = 2) = \binom{10}{2} 0{,}3^2 0{,}7^8$, $P(X = 0) = 0{,}7^{10}$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{30}{10}$.

Reference in the manual

Supplements: Calao, p. 164, Nos. 24 to 29, then p. 166, No. 36.
is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 161, exercise 13

We throw a well-balanced coin five times in a row. F the random variable which, at each experience thus defined, combines the number of "faces" obtained.

1. F follow a binomial distribution, specify its parameters.
2. Realize the triangle of Pascal up to $n = 5$.
3. Calculate $P(F = 3)$. Interpret this result.
4. Calculate the probability that the "Face" side will appear at least three times.

Detailed correction · Professor edition

$F \sim \mathcal{B}(5; 1/2)$. The line $n = 5$ of the triangle is 1, 5, 10, 10, 5, 1. $P(F = 3) = \binom{5}{3}(1/2)^5 = 10/32 = 0.3125$ and $P(F \geqslant 3) = (10 + 5 + 1)/32 = 0{,}5$.

Personal work

Resuming Exercise 2 with the following data:
365 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 50 — Binomial probabilities

Discrete random variables

29/04/2027 • B : 1 h – C : 2 h

Objectives of the session

· Calculate $P(X = k)$.

· Translate at least/at most.

· Use the supplement.

Fluency · 6 to 8 minutes, without calculator

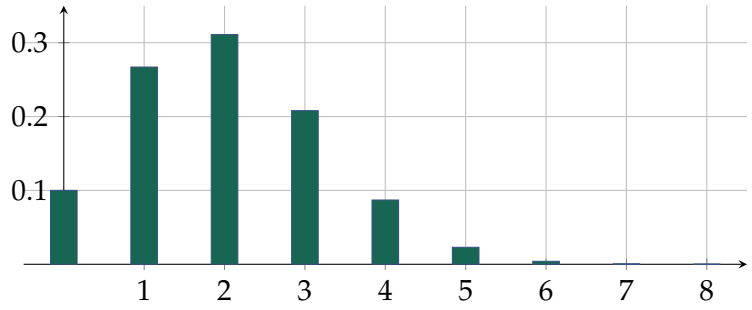
1. Reduce 3 $x + 4 - x + 2$. $2x + 6$
2. Translate $X \geq 1$ by a supplementary. $1 - P(X = 0)$
3. Provide $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3
4. Calculate $\binom{5}{2}$. 10
5. Calculating the Expectation of X constant equal to 4. 4

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: calculate $p(x = k)$ and translate at least/at most.

Research activity

For $X \sim \mathcal{B}(8, 0{,}25)$, calculate $P(X = 2)$, $P(X = 0)$ then $P(X \geq 1)$. Translate each event into sentence.



Detailed correction · Professor edition

$P(X = 2) = \binom{8}{2} 0{,}25^2 0{,}75^6 \approx 0{,}3115$; $P(X = 0) = 0{,}75^8 \approx 0{,}1001$; $P(X \geq 1) \approx 0{,}8999$. Make verbalize the data that triggers the choice of method.

Trace written

If $X \sim \mathcal{B}(n, p)$, then $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$. Carefully translate "at least", "at most and "none".

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$P(X=2)=\binom{8}{2}0{,}25^2 0{,}75^6 \approx 0{,}3115$; $P(X=0)=0{,}75^8 \approx 0{,}1001$; $P(X\geq 1)\approx 0{,}8999$.

Training · essential

1. Justify that the number of successes among 6 independent trials, with $p=0{,}3$, follows a law binomial.
2. Calculate $P(X=2)$ and $P(X=0)$.
3. Deduct from it $P(X\geq 1)$ and calculate $E(X)$.

Detailed correction · Professor edition

$X\sim\mathcal{B}(6,0{,}3)$. $P(X=2)=\binom{6}{2}0{,}3^2 0{,}7^4$, $P(X=0)=0{,}7^6$, $P(X\geq 1)=1-P(X=0)$ and $E(X)=\dfrac{18}{10}$.

Reference in the manual

Supplements: Calao, p. 164, Nos. 24 to 29, and p. 168, Nos. 41 to 45.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 167, exercise 38

A parcel delivery company succeeds, for packages taken care of before 10 a.m., in delivering them as soon as possible the next day with a probability of 0 ,75. A merchant shall deposit before 10:07 a.m. parcels for his Clients. X the random variable which, in each batch of seven packages, combines the number of packages The next day, we admit that X follows a binomial distribution.

1. Determine parameters n and p of the law X.
2. Calculate the probability of the following events: A: "All packages will arrive the next day." ; B: "At least one package will not arrive the next day."
3. Complete Pascal's triangle up to the line $n=7$. Deduct $\binom{7}{2}$. Calculate probability Event C: " Exactly two packages will arrive the next day."
4. Interpret the event $\{X=4\}$. Calculate probability $P(X=4)$.

Detailed correction · Professor edition

$X\sim\mathcal{B}(7;0{,}75)$. $P(A)=0{,}75^7\approx 0{,}1335$ and $P(B)=1-0{,}75^7\approx 0{,}8665$. $\binom{7}{2}=21$, $P(X=2)=21(0{,}75)^2(0{,}25)^5\approx 0{,}0115$. $\{X=4\}$ means that exactly four packages are delivered on tomorrow and $P(X=4)=\binom{7}{4}(0{,}75)^4(0{,}25)^3\approx 0{,}1730$.

Personal work

Resuming Exercise 3 with the following data:
370 and the rate or common difference / ratio by 4. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 51 — Chapter review and evaluation

Discrete random variables

04/05/2027 • B : 2 h – C : 1 h

Objectives of the session

· Calculate and interpret $E(X)$.

· Solve a binomial problem.

· Analyze errors.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 100. 20

2. Provide $E(X)$ for $X \sim \mathcal{B}(10, 0{,}3)$. 3

3. Calculate $\binom{5}{2}$. 10

4. Calculate the expectation of X constant equal to 5. 5

5. Translate $X \geq 1$ by a supplementary. $1 - P(X = 0)$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: calculate and interpret $e(x)$ and solve a problem binomial.

Research activity

One centre receives 120 groups of 10 clients, each client probably claiming 0 ,08. Estimate the average number of complaints per group, then on all groups.
Expectation may or may not predict.

Detailed correction · Professor edition

By group $E(X) = 10 \times 0{,}08 = 0{,}8$. Out of 120 groups, an average of approximately 96 claims; this is not a certainty. Make verbalize the data that triggers the choice of method.

Trace written

For $X \sim \mathcal{B}(n, p)$, $E(X) = np$. The value gives the average number of successes over a large number of series of n tests.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

By group $E(X) = 10 \times 0{,}08 = 0{,}8$. Out of 120 groups, an average of approximately 96 claims; this is not a certainty.

Training · essential

1. Justify that the number of successes among 7 independent trials, with $p = 0{,}3$, follows a law binomial.
2. Calculate $P(X = 2)$ and $P(X = 0)$.
3. Deduct from it $P(X \geq 1)$ and calculate $E(X)$.

Detailed correction · Professor edition

$X \sim \mathcal{B}(7, 0{,}3)$. $P(X = 2) = \binom{7}{2} 0{,}3^2 0{,}7^5$, $P(X = 0) = 0{,}7^7$, $P(X \geq 1) = 1 - P(X = 0)$ and $E(X) = \dfrac{21}{10}$.

Reference in the manual

Supplements: Calao, pp. 160-169. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 166, exercise 34

A company markets, in large quantities, low consumption and long-lasting LED bulbs 98% of these bulbs have a running time of at least 20,000 hours. bulbs are sold in lots of 10. X the random variable which, in each batch of 10 ampoules, combines the number of bulbs whose operating time is at least equal to 20 000 h. The quantity of ampoules is large enough to assimilate the choice of a batch at random sampling and with 10 ampoules delivered.

1. What is the law followed by X ? Specify its parameters.
2. Copy and complete: $\binom{10}{9} = \cdots$; $\binom{10}{10} = \cdots$.
3. Deduce probabilities (round to 0 ,001) : $P(X = 9)$ and $P(X = 10)$. Interpret results previous.
4. Deduce the probability (at 0 ,001) that a batch of 10 ampoules contains at least 9 bulbs with a running time of at least 20,000 hours.

Detailed correction · Professor edition

$X \sim \mathcal{B}(10; 0{,}98)$, $\binom{10}{9} = 10$ and $\binom{10}{10} = 1$. $P(X = 9) = 10(0{,}98)^9(0{,}02) \approx 0{,}167$ and $P(X = 10) = 0{,}98^{10} \approx 0.817$. $P(X \geqslant 9) \approx 0{,}984$.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.